

```
1: #include<stdio.h>
2: int main()
3: {
4:     int i,j,r,c,trace=0;
5:     printf("Enter the number of rows and columns:\n");
6:     scanf("%d%d",&r,&c);
7:     int tra_arr[r][c];
8:     printf("Enter the elements of a matrix:\n");
9:     for(i=0;i<r;i++)
10:    {
11:        for(j=0;j<c;j++)
12:        {
13:            scanf("%d",&tra_arr[i][j]);
14:        }
15:    }
16:    for(i=0;i<r;i++)
17:    {
18:        for(j=0;j<c;j++)
19:        {
20:            if(i==j)
21:            {
22:                trace=trace+tra_arr[i][j];
23:            }
24:        }
25:    }
26:    printf("the trace of a matrix is %d\n",trace);
27: }
```